Practical No. 3: Jenkins Pipeline to Pull Docker Image & Run Container

Objective

To create a Jenkins pipeline that pulls a Docker image from Docker Hub and runs it as a container.

Prerequisites

1. Jenkins installed and running
2. Docker Desktop installed and running
3. Git installed
4. Basic knowledge of Jenkins pipeline

Description

This pipeline demonstrates integration of Jenkins with Docker by pulling an image from Docker Hub and running it as a container. It includes stages to verify Docker installation, pull the image, run the container, and validate execution.

Steps

1. Verify Docker installation using docker --version and docker ps
2. Ensure Docker is added to system PATH
3. Create a Jenkins pipeline job named docker-pipeline
4. Add pipeline script
5. Run the pipeline
6. Verify output in browser (http://localhost:8085)

Pipeline Script

```
pipeline {
  agent any

  stages {
    stage('Check Docker') {
      steps {
        bat 'docker --version'
      }
    }
```

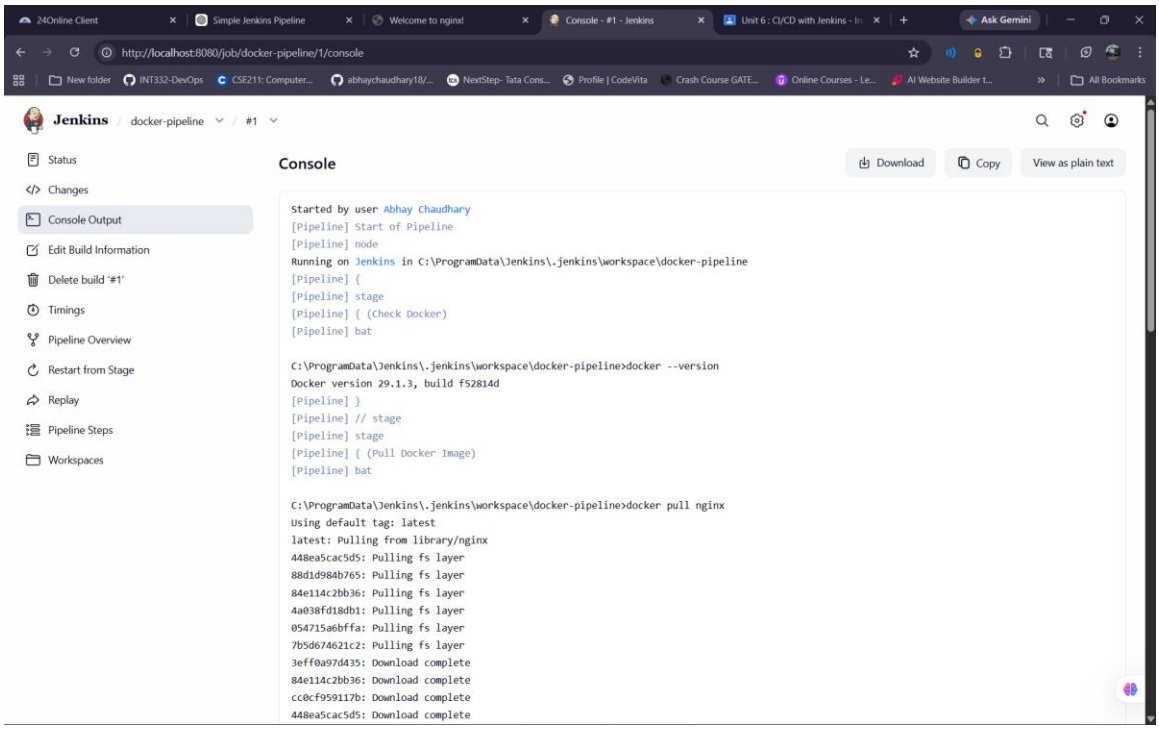
```
    stage('Pull Docker Image') {
      steps {
        bat 'docker pull nginx'
      }
    }

    stage('Remove Old Container') {
      steps {
        bat 'docker rm -f my-nginx || exit 0'
      }
    }

    stage('Run Container') {
      steps {
        bat 'docker run -d -p 8085:80 --name my-nginx nginx'
      }
    }

    stage('Verify Container') {
      steps {
        bat 'docker ps'
      }
    }
  }
}
```

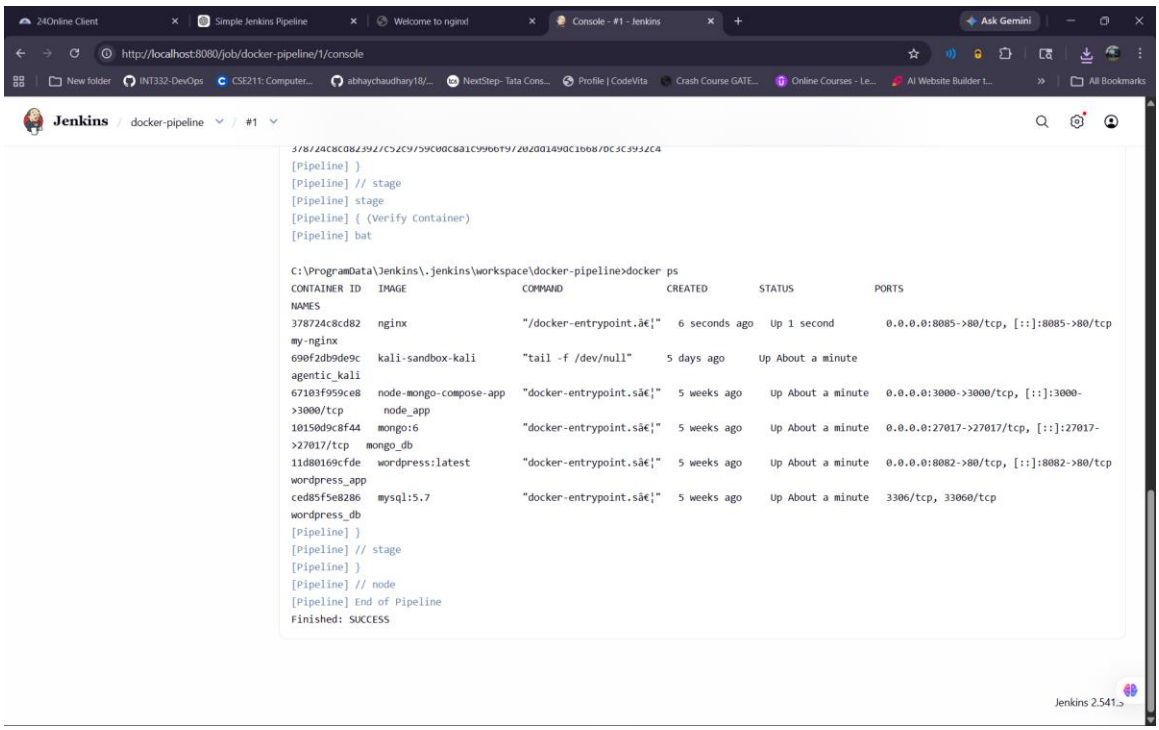
Expected Output

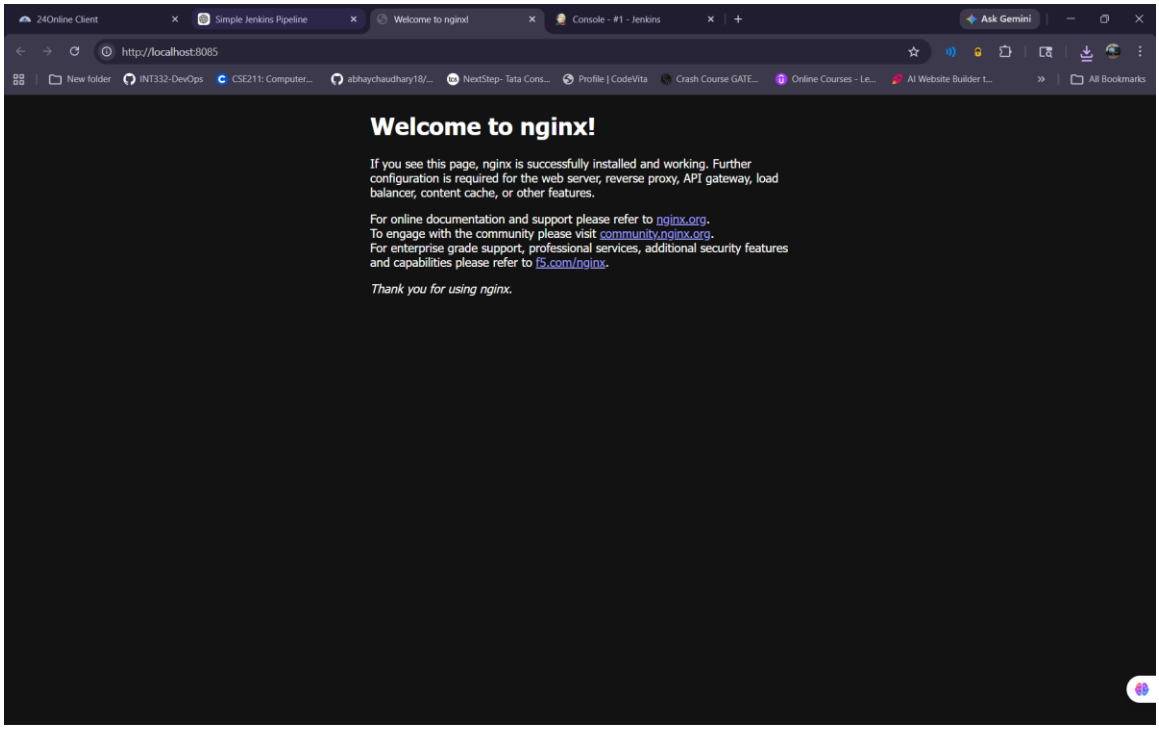
Docker image pulled successfully
Container started successfully
docker ps shows running container
NGINX welcome page visible at http://localhost:8085
Finished: SUCCESS

Conclusion

Successfully implemented a Jenkins pipeline to pull a Docker image and run it as a container. The container was verified using browser and Docker commands.


24Online Client
Simple Jenkins Pipeline
Welcome to nginx!
Console - #1 - Jenkins
Unit 6 : CI/CD with Jenkins - In
Ask Gemini
http://localhost:8080/job/docker-pipeline/1/console
New folder
INT332-DevOps
CSE211: Computer...
abhaychaudhary18/...
NextStep- Tata Cons...
Profile | CodeVita
Crash Course GATE...
Online Courses - Le...
AI Website Builder t...
All Bookmarks
Jenkins / docker-pipeline / #1
Status
Changes
Console Output
Edit Build Information
Delete build '#1'
Timings
Pipeline Overview
Restart from Stage
Replay
Pipeline Steps
Workspaces
Console
Download
Copy
View as plain text
Started by user Abhay Chaudhary
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\docker-pipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Check Docker)
[Pipeline] bat
C:\ProgramData\Jenkins\.jenkins\workspace\docker-pipeline>docker --version
Docker version 29.1.3, build f52814d
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Pull Docker Image)
[Pipeline] bat
C:\ProgramData\Jenkins\.jenkins\workspace\docker-pipeline>docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
448ea5cac5d5: Pulling fs layer
88d1d984b765: Pulling fs layer
84e114c2bb36: Pulling fs layer
4a038fd18db1: Pulling fs layer
054715a6bffa: Pulling fs layer
7b5d674621c2: Pulling fs layer
3eff0a97d435: Download complete
84e114c2bb36: Download complete
cc0cf959117b: Download complete
448ea5cac5d5: Download complete


24Online Client
Simple Jenkins Pipeline
Welcome to nginx!
Console - #1 - Jenkins
Ask Gemini
http://localhost:8080/job/docker-pipeline/1/console
New folder
INT332-DevOps
CSE211: Computer...
abhaychaudhary18/...
NextStep- Tata Cons...
Profile | CodeVita
Crash Course GATE...
Online Courses - Le...
AI Website Builder t...
All Bookmarks
Jenkins / docker-pipeline / #1
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Verify Container)
[Pipeline] bat
C:\ProgramData\Jenkins\.jenkins\workspace\docker-pipeline>docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS
NAMES
378724c8cd82 nginx "/docker-entrypoint.â€¦" 6 seconds ago Up 1 second 0.0.0.0:8085->80/tcp, [::]:8085->80/tcp
my-nginx
690f2db9de9c kali-sandbox-kali "tail -f /dev/null" 5 days ago Up About a minute
agentic_kali
67103f959ce8 node-mongo-compose-app "docker-entrypoint.sâ€¦" 5 weeks ago Up About a minute 0.0.0.0:3000->3000/tcp, [::]:3000-
>3000/tcp node_app
10150d9c8f44 mongo:6 "docker-entrypoint.sâ€¦" 5 weeks ago Up About a minute 0.0.0.0:27017->27017/tcp, [::]:27017-
>27017/tcp mongo_db
11d80169cfde wordpress:latest "docker-entrypoint.sâ€¦" 5 weeks ago Up About a minute 0.0.0.0:8082->80/tcp, [::]:8082->80/tcp
wordpress_app
ced85f5e8286 mysql:5.7 "docker-entrypoint.sâ€¦" 5 weeks ago Up About a minute 3306/tcp, 33060/tcp
wordpress_db
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
Jenkins 2.541.


24Online Client
Simple Jenkins Pipeline
Welcome to nginx!
Console - #1 - Jenkins
Ask Gemini
http://localhost:8085
New folder
INT332-DevOps
CSE211: Computer...
abhaychaudhary18/...
NextStep- Tata Cons...
Profile | CodeVita
Crash Course GATE...
Online Courses - Le...
AI Website Builder t...
All Bookmarks
Welcome to nginx!
If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.
For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.
Thank you for using nginx.